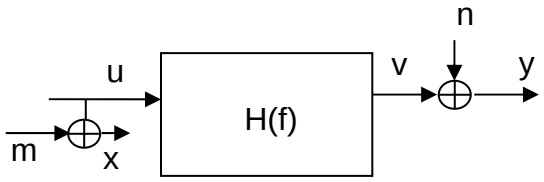
Problem 10.1 (MATLAB)
System Identification (FRF, Frequency Response Functions)

In this problem is studied methods for system identification in terms of estimating the Frequency Response Function (FRF) for an unknown LTI-system based on (potentially) noisy measurements of the input and output signals to the LTI-system.
The general block diagram for the problem is shown below:



The signals are

- u: true input signal
- m: input measurement noise
- x = u + m: measured input signal
- v: true output signal
- n: output measurement noise
- y = v + n: measured output signal

This general set-up will be studied in 4 different cases, using the FRF estimators

$$\hat{H}_1(f) = \frac{\hat{G}_{xy}(f)}{\hat{G}_{xx}(f)} \quad \text{and}$$

$$\hat{H}_2(f) = \frac{\hat{G}_{yy}(f)}{G^*_{xy}(f)}$$

Also studied will be the (squared) coherence function

$$\gamma^2_{xy}(f) \stackrel{\text{def}}{=} \frac{|\hat{G}_{xy}(f)|^2}{\hat{G}_{xx}(f)\hat{G}_{yy}(f)}$$

which measures to what degree the measured ouput signal is generated by linear filtering of the measured input signal.

System set-up

- Define an “unknown” (discrete-time) LTI-system, i.e. define an even number of complex conjugate poles (peaks in H(f)) and zeros (valleys in H(f)), e.g. 4 poles and 2 zeroes giving 2 peaks and 1 valley
- Plot the true amplitude and phase characteristics for H(f) and adjust pole and zero positions in z-plane until a satisfactory H(f) is achieved
- As input signal, u, can be used white noise or other signal broadband enough to “cover” the frequency range where |H(f)| is not very small
- As input measurement noises, m and n, can be used white noise signals
- In the following, start with relative high signal lengths, e.g. N = 4096 or more, then afterwards try with smaller N, e.g. N = 256 or N = 512

Case 1: <u>no input or output noise, m = n = 0</u>

- **Generate x = u and filter u with H(f) to achieve v = y**
- Calculate the H_1 and H_2 estimators using either *pwelch/cpsd* or *tfestimate* and plot their FRF's
- Do this a number of times using overlay plots with e.g. 10-50 overlays
- Plot the H_1 and H_2 mean FRF's and compare with the true H(f)
- Calculate also the (squared) coherence function using *pwelch/cpsd* or *mscohere* and plot it together with the theoretical version
- Comment
- Use different signal block lengths and comment

Case 2: <u>only output noise, m = 0</u>

- **Generate x = u and filter u with H(f) to achieve v, then add noise: y = v + n**
- Repeat the steps from Case 1

Case 3: <u>only input noise, n = 0</u>

- **Generate u, add noise to achieve x = u + m, filter u (!) with H(f) to achieve v = y**
- Repeat the steps from Case 1

Case 4: <u>both input and output noise</u>

- **Generate u, add noise to achieve x = u + m, filter u (!) with H(f) to achieve v, add noise to achieve y = v + n**
- Repeat the steps from Case 1
- Calculate and plot also as a compromise between H_1 and H_2 their geometric mean, e.g. the square root of their product